

# ETHAN VILLALOVOZ

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## EDUCATION

### Georgia Institute of Technology

Jan 2026 – May 2028

M.S. in Computer Science (Computational Perception & Robotics) — GPA: 4.0/4.0

Atlanta, GA

### Washington State University, Honors College

Aug 2021 – May 2025

B.S. in Computer Science, Minor in Mathematics — GPA: 3.94/4.0, Summa Cum Laude

Pullman, WA

## EXPERIENCE

### Microsoft

May 2026 – Present

Software Engineer Intern

Redmond, WA

- Building a **multi-agent workflow system** that integrates agent orchestration into an internal cloud platform

### Georgia Institute of Technology

Feb 2026 – Present

Graduate Student Researcher

Atlanta, GA

- Developing a **video-to-action robot policy framework** on **WAN 2.2**, combining video diffusion, action flow matching, and **PyTorch/Hydra** pipelines for **LIBERO** rollouts

### Washington State University

Jan 2024 – May 2025

Undergraduate Research Assistant

Pullman, WA

- Built a test-driven **LLM code-generation** evaluation pipeline, measuring functional correctness on **164 HumanEval+** Python tasks by executing generated programs across three code LLMs
- Validated **BODE-GEN** gains with reproducible benchmarks: **89%** vs. **61–78%** on ChatGPT-3.5, **50%** vs. **22–38%** on CodeLlama-7B, and **94%** vs. **66–78%** on DeepSeek-Coder-33B

### Carnegie Mellon University

Jun 2024 – Aug 2024

Robotics Institute Summer Scholar

Pittsburgh, PA

- Developed a **Bayesian reward-learning** system for a simulated dishwasher-loading robot, modeling preferences as **hierarchical decision-tree rewards** over object and placement attributes
- Implemented **state-correction** and **feature-clarification** loops over **12 candidate preference models**, using simulated rollouts to identify the correct reward hypothesis

### Google

May 2023 – Aug 2023

Software Engineering Intern (STEP)

Sunnyvale, CA

- Developed and deployed **5 C++/SQL analytics jobs** to compute internal database queue metrics, replacing manual analysis with automated reporting for engineering stakeholders
- Scaled metric computation from **1% to 100%** dataset coverage within **4 hours**, reducing runtime by **66%**; built internal **HTML/SQL dashboards** with live statistical summaries

## PUBLICATIONS

**An Exploratory Study of Bayesian Prompt Optimization for Test-Driven Code Generation with Large Language Models.** S. Tomar, A. Deshwal, **E. Villalovoz**, M. Fazzini, H. Cai, J.R. Doppa. *arXiv:2512.15076*, 2025.

**Social Triangles and Aggressive Lines: Multi-Robot Formations Impact Navigation and Approach.** A. Bacula, **E. Villalovoz**, D. Flynn, A. Mehta, H. Knight. *IROS*, 2023.

## PROJECTS

**Knowledge Graph RAG Assistant** | React/TS, FastAPI, FAISS, DBpedia, OpenAI, Docker

- Contributed to the retrieval stack for a full-stack **RAG application**, combining FastAPI, FAISS vector search, DBpedia/SPARQL context, and OpenAI responses; generated **5GB+** Wikipedia embeddings for **10,000+** articles

## TECHNICAL SKILLS

**Languages:** Python, C/C++, SQL, TypeScript, JavaScript, HTML/CSS, C#, MATLAB, Swift, GLSL

**AI/ML & Robotics:** PyTorch, TensorFlow, scikit-learn, LightGBM, Hugging Face Transformers, FAISS, ROS, OpenCV

**Web/Backend:** React, Next.js, FastAPI, Flask, REST APIs, SQLite, MySQL, DBpedia/SPARQL

**Developer Tools:** Git, GitHub Actions, Docker, AWS, Bash, Conda, Jupyter, MLflow, DVC, Xcode